FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Common Cancers in Children

Tutorial | Week 5 · Measles, vaccines & the newborn | Paediatrics | Dr Nabatte

Childhood cancers in Uganda (Burkitt lymphoma, Kaposi sarcoma, leukaemia, Wilms tumour, retinoblastoma, neuroblastoma) are highly curable when caught early, yet most children present late; your job is to recognise the warning signs, avoid harmful delays (steroids, repeated tumour palpation, unnecessary surgery), prevent tumour lysis, and refer quickly to the Uganda Cancer Institute.

Must know before the ward round

1. WHO childhood cancer fact sheet: about 400,000 children and adolescents aged 0–19 get cancer each year. More than 80% are cured in high-income countries but less than 30% in many low- and middle-income countries, mainly because of late diagnosis and abandoned treatment.
2. Burkitt lymphoma is the classic East African childhood cancer: an EBV- and malaria-linked B-cell lymphoma with a MYC translocation t(8;14). It doubles in about 25 hours and presents as a jaw or abdominal mass at 4–7 years, more often in boys.
3. Wilms tumour: a smooth, firm flank mass in a well child aged 2–5 years that does NOT cross the midline; haematuria and hypertension in about a quarter. Palpate gently and only once (rupture upstages it).
4. Neuroblastoma: under 5 years (median about 18 months), an irregular hard mass that crosses the midline in a sick child, with periorbital bruising, bone pain, raised urinary catecholamines (VMA/HVA) and calcification on imaging.
5. Retinoblastoma: leukocoria (white pupil) or a new squint in a child under 5, RB1 gene on chromosome 13q14. Bilateral disease is heritable. UCG 2023: refer every suspected case to an ophthalmologist within 72 hours.
6. ALL is the commonest childhood leukaemia (about 75–80%), peaking at 2–5 years: pallor, fever, bleeding, bone pain or limp, hepatosplenomegaly and nodes. Never give steroids before the diagnosis is made.
7. Paediatric Kaposi sarcoma in Uganda is often HIV-associated. It presents with lymph node (lymphadenopathic) disease, woody oedema or palate lesions more than skin lesions. Treat with ART plus chemotherapy at UCI.
8. Tumour lysis syndrome (Cairo–Bishop) is 2 or more of: uric acid ≥476 µmol/L, K^+ ≥6.0 mmol/L, phosphate ≥2.1 mmol/L (children), Ca^{2+} ≤1.75 mmol/L, or a 25% change from baseline, between 3 days before and 7 days after treatment.
9. TLS prevention: hyperhydration 2–3 L/m²/day (no potassium), urine output 80–100 mL/m²/h, and allopurinol 10 mg/kg/day in 3 doses (max 800 mg/day) from 24–48 h before chemotherapy. Rasburicase 0.2 mg/kg is the alternative for high risk but is contraindicated in G6PD deficiency.
10. Febrile neutropenia (ANC <0.5 × 10^9/L and temperature ≥38.3 °C once or ≥38.0 °C for 1 h) is an emergency: cultures, then broad-spectrum IV antibiotics within 60 minutes.
11. Early warning signs follow SILUAN: Seek help early, Insight (white eye, squint, proptosis), Lump, Unexplained fever/weight loss/pallor/bruising, Aching bones, Neurological signs.
12. Uganda pathway: HC → general hospital → regional referral (FBC, ultrasound, FNA) → Uganda Cancer Institute paediatric unit (Mulago). Chemotherapy there is free; treat pain with WHO 2-step analgesia including oral morphine.

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1. Definition

Childhood cancer is malignancy arising in children and adolescents (0–19 years in the WHO fact sheet; UCG 2023 counts children 0–14 years). Cancer is **rare in children compared with adults**, but it is a leading cause of death from disease in children worldwide. The pattern in Uganda differs from Europe: **Burkitt lymphoma** and **Kaposi sarcoma** are common here, alongside leukaemia, Wilms tumour, retinoblastoma, Hodgkin lymphoma and sarcomas.

Most childhood cancers are **very sensitive to chemotherapy** and are **curable**.

Key terms

Term	Definition
Staging	How far the tumour has spread (e.g. Murphy/St Jude stages I–IV for Burkitt; SIOP/COG stages I–V for Wilms; INRGSS for neuroblastoma)
Febrile neutropenia	Absolute neutrophil count (ANC) **<0.5 × 10^9/L** (or expected to fall below this) with a temperature **≥38.3 °C once or ≥38.0 °C sustained for 1 hour**
Tumour lysis syndrome (TLS)	Metabolic chaos when many tumour cells die at once (spontaneously or after chemotherapy), releasing **potassium, phosphate and nucleic acids (which become uric acid)**
Abandonment of treatment	Missing **4 or more consecutive weeks** of planned curative therapy; a leading cause of treatment failure in Uganda
Leukocoria	A white pupillary reflex instead of red: retinoblastoma until proven otherwise
Hyperleukocytosis	WBC **>100 × 10^9/L**: risk of stroke, respiratory failure and TLS

> CLINICAL PEARL
> The one thing never to forget: **a child with a lump, a white pupil, unexplained bruising or pallor, persistent bone pain or a rapidly growing jaw swelling needs a diagnosis this week, not after another course of antibiotics or dewormers.** Your fast referral is the first step of the cure.

2. Epidemiology

- **Global (WHO fact sheet)**: about **400,000 children and adolescents aged 0–19** get cancer each year. **Over 80%** survive in high-income countries but **under 30%** in many low- and middle-income countries. The difference comes from:
  1. **Late diagnosis**: a "toothache" treated with extractions for 2 months, an eye "infection" for 6 months, a "big tummy from worms".
  2. **Abandonment**: families stop treatment because of cost of transport and food, fear, or traditional healers.

3. **Treatment-related deaths**: infection, tumour lysis and malnutrition.

- **WHO Global Initiative for Childhood Cancer (GICC, "CureAll")**: aims for **at least 60% survival by 2030**, focusing on six index cancers: **ALL, Burkitt lymphoma, Hodgkin lymphoma, retinoblastoma, Wilms tumour and low-grade glioma**.
- **Uganda**: children (0–14 years) make up about **10% of cancer patients** (UCG 2023). The **Kampala Cancer Registry** and the **Uganda Cancer Institute (UCI)** show that **Burkitt lymphoma, Kaposi sarcoma, leukaemias, Wilms tumour, retinoblastoma and Hodgkin lymphoma** account for most childhood cancers seen in Uganda.
- **History**: **Burkitt lymphoma was first described in Kampala**. In 1958 the Irish surgeon **Denis Burkitt**, working at Mulago, mapped a "lymphoma belt" across malaria-endemic equatorial Africa. Epstein and colleagues later found the **Epstein–Barr virus (EBV)** in his tumour samples, the first virus linked to a human cancer.
- **Kaposi sarcoma** rose with the HIV epidemic. With better ART coverage and **early infant HIV diagnosis**, it is falling but still seen.
- **Services**: the **UCI paediatric oncology unit (Mulago, Kampala)** is the national centre, with free chemotherapy, a children's ward and outpatient day care. **Lacor Hospital (Gulu)** has a long-running Burkitt lymphoma programme. Regional cancer services are being developed (e.g. **Gulu and Mbarara**) Verify *current status*. **Mulago Eye** (Department of Ophthalmology) handles retinoblastoma. Radiotherapy is only at UCI. **Palliative care**: Uganda was a pioneer of **oral morphine for palliative care** (Hospice Africa Uganda), and it is available free through the public system.
- **Barriers**: travel costs to Kampala, food and lodging for caretakers, stigma ("cancer is a death sentence"), use of traditional healers, stock-outs, and limited pathology outside Kampala.

> **UGANDA CONTEXT**
> At a HC III/IV you will not confirm cancer, but you **can** suspect it, do a **FBC, malaria test, HIV test and abdominal ultrasound** where available, start **allopurinol and hydration** for a big Burkitt or leukaemia, treat infection and pain, and refer **urgently with a clear letter** to UCI (via the regional referral hospital if needed). Tell the family that **treatment at UCI is free** and that many children are cured; this helps prevent abandonment.

3. Aetiology and Risk Factors

Most childhood cancers arise from **embryonal cells** that fail to mature (e.g. nephrogenic rests → Wilms; neural crest → neuroblastoma; retinal cells → retinoblastoma) or from **lymphoid cells** during rapid immune development. Unlike adult cancers, they are **rarely caused by lifestyle**. Known contributors:

Factor	Cancer
EBV + holoendemic malaria	Endemic Burkitt lymphoma. Malaria drives B-cell proliferation and weakens T-cell control of EBV, giving more chances for the **MYC** translocation
HIV + HHV-8 (KSHV)	Kaposi sarcoma; HIV also increases NHL (including Burkitt)
RB1 mutation (chromosome 13q14)	Retinoblastoma (Knudson "two-hit" hypothesis); also later osteosarcoma. Bilateral disease is heritable
WT1 (11p13), syndromes	Wilms: WAGR (Wilms, aniridia, genitourinary anomalies, retardation), Denys–Drash, Beckwith–Wiedemann (also hepatoblastoma), hemihypertrophy
Down syndrome	ALL (10–20× risk) and AML (especially megakaryoblastic)
Li–Fraumeni (TP53), NF1, ionising radiation, prior chemotherapy	Various

Risk factors for tumour lysis syndrome: **Burkitt lymphoma and ALL with high WBC (>100 × 10^9/L)**, bulky disease, high LDH (>2× normal), pre-existing renal impairment and dehydration.

Risk factors for abandonment and poor outcome: distance from Kampala, cost of transport and food, no caretaker able to stay, stigma, traditional healers, malnutrition.

4. Pathophysiology

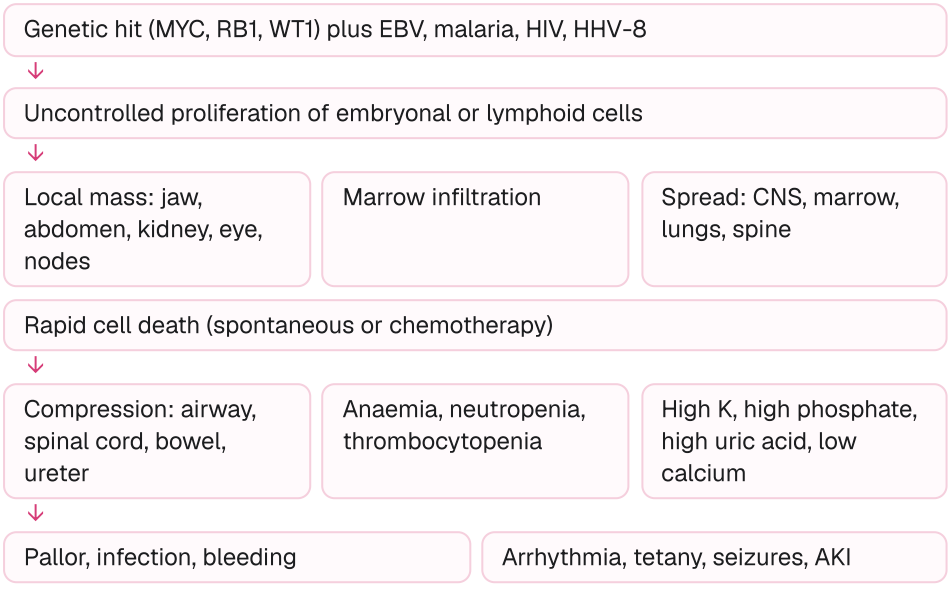
Burkitt lymphoma: the fastest human tumour

- A B-cell lymphoma with **MYC on chromosome 8** translocated next to an immunoglobulin gene: **t(8;14)** (about 80%), or t(2;8) or t(8;22). MYC drives constant cell division.
- **Growth fraction close to 100% and a doubling time of about 25 hours**. A jaw mass can double in a day or two. This explains both its **high chemosensitivity** (cures with fairly simple regimens) and its **high risk of tumour lysis**.
- **Histology**: "**starry sky**" pattern (sheets of blue tumour cells with pale macrophages eating dead cells), **Ki-67 ~100%**, CD20+, CD10+, BCL2–.
- **Three types**: **endemic** (Africa; jaw, orbit, abdomen, kidneys, ovaries; almost 100% EBV+), **sporadic** (Western; abdomen, ileocaecal; about 20% EBV+), **immunodeficiency-associated** (HIV, transplant).

Tumour lysis syndrome: when cells die, they release:

- **Potassium** → hyperkalaemia → arrhythmias, cardiac arrest.
- **Phosphate** → hyperphosphataemia → binds calcium → **hypocalcaemia** (tetany, seizures) and **calcium phosphate crystals in the kidney** → AKI.
- **Nucleic acids** → purines → hypoxanthine → xanthine → **uric acid** (via xanthine oxidase) → **uric acid crystals** in renal tubules → AKI.

Allopurinol blocks xanthine oxidase (stops new uric acid). **Rasburicase** converts existing uric acid to soluble allantoin.



Feature links

- Rapidly dividing B cells in the jaw → **jaw or maxillary swelling, loose teeth**, proptosis (orbit).
- Abdominal Burkitt → **multiple masses, ascites**, kidney and ovary involvement; ileocaecal disease → **intussusception or bowel obstruction** in a child over 3.
- Renal embryonal tumour confined by the capsule → **smooth flank mass that does not cross the midline**; renin release → **hypertension**; collecting system invasion → **haematuria**.

- Neural crest tumour → **irregular, calcified mass crossing the midline**; catecholamines → **hypertension, raised VMA/HVA**; orbital metastases → **periorbital bruising ("raccoon eyes")**, proptosis; marrow and bone → **bone pain**; skin → "blueberry muffin" nodules in infants.
- Retinal tumour reflects light → **leukocoria**; loss of central vision → **squint**.
- Marrow replacement → **pallor** (anaemia), **fever and infections** (neutropenia), **petechiae, bruising, nosebleeds** (thrombocytopenia), **bone pain and limp**.
- Lymphoid infiltration → **hepatosplenomegaly, lymphadenopathy**, testicular enlargement.
- Anterior mediastinal mass (T-cell ALL/lymphoma) → **stridor, orthopnoea, facial swelling, distended neck veins** (superior mediastinal syndrome).
- Paraspinal or epidural tumour → **back pain, leg weakness, urinary retention** (cord compression).
- HHV-8 endothelial proliferation → **purple skin or palate lesions, woody oedema, lymphadenopathy** (Kaposi sarcoma).
- Cytokines and tumour burden → **fever, night sweats, weight loss** (B symptoms), **high LDH**.
- Mass in the posterior fossa → **morning headache, vomiting, ataxia**, enlarging head (brain tumour).

5. Classification / Staging

Groups of childhood cancer

Group	Examples	Typical age
Haematological	Acute lymphoblastic leukaemia (ALL), acute myeloid leukaemia (AML), chronic myeloid leukaemia (rare)	ALL 2–5 y; AML any age
Lymphomas	**Burkitt lymphoma** (non-Hodgkin), Hodgkin lymphoma, other NHL	Burkitt 4–7 y; Hodgkin older children/teens
Embryonal tumours ("-blastomas")	**Wilms tumour** (nephroblastoma), **neuroblastoma**, **retinoblastoma**, hepatoblastoma, medulloblastoma	Mostly under 5
Sarcomas	Rhabdomyosarcoma, osteosarcoma, Ewing sarcoma, **Kaposi sarcoma**	Rhabdo any age; bone sarcomas teens
CNS tumours	Low-grade glioma, medulloblastoma, ependymoma, craniopharyngioma	Any age
Germ cell tumours	Sacrococcygeal teratoma, testicular, ovarian	Infants and teens

Key facts by tumour

Cancer	Age and sex	Hallmark
Burkitt lymphoma	4–7 years (UCG), boys > girls	Jaw or abdominal mass; t(8;14); doubles in ~25 h
ALL	Peak 2–5 years; ~75–80% of childhood leukaemia	Pallor, fever, bleeding, bone pain or limp, hepatosplenomegaly, nodes
Wilms tumour	2–5 years (UCG average 2)	Smooth flank mass in a well child; haematuria and hypertension in about a quarter
Neuroblastoma	Under 5 years; median ~18 months	Irregular mass crossing the midline in a sick child; raised VMA/HVA in >90%
Retinoblastoma	Usually under 3–5 years	Leukocoria, squint; RB1 13q14
Kaposi sarcoma	Any age; often HIV-associated	Lymphadenopathic disease, woody oedema, palate lesions more than skin lesions

Cancer	Age and sex	Hallmark
Hodgkin lymphoma	Older children and teens	Painless rubbery cervical nodes; Reed–Sternberg cells

Burkitt lymphoma staging (Murphy/St Jude)

Stage	Disease
I	Single tumour (extranodal) or single nodal area, **excluding mediastinum and abdomen**
II	Single extranodal tumour with regional nodes; ≥2 nodal areas on the **same side** of the diaphragm; resectable primary GI tumour
III	Disease on **both sides** of the diaphragm; any **primary intrathoracic** tumour; **extensive intra-abdominal** disease; **paraspinal or epidural** tumour
IV	**CNS and/or bone marrow** involvement

A jaw-only tumour is stage I; jaw plus extensive abdominal disease is III; any CSF or marrow involvement is IV.

Tumour lysis syndrome (Cairo–Bishop): laboratory TLS = 2 or more of these, from 3 days before to 7 days after starting therapy.

Parameter	Threshold (or 25% change from baseline)
Uric acid	≥476 µmol/L (8 mg/dL)
Potassium	≥6.0 mmol/L
Phosphate	≥2.1 mmol/L in children (≥1.45 in adults)
Calcium	≤1.75 mmol/L (25% decrease)

Clinical TLS = laboratory TLS plus one of: **creatinine ≥1.5 × upper limit of normal**, **arrhythmia/sudden death**, or **seizure**.

6. Clinical Features

History

Ask about	Why it matters
Duration and speed of growth of a lump	Burkitt grows over days to weeks; Wilms over weeks; benign lumps slowly or not at all
Loose teeth, "toothache", dental extractions, facial swelling	Classic endemic Burkitt (UCG: extractions do not relieve the swelling)
Abdominal distension, pain, vomiting, constipation, urinary retention	Abdominal Burkitt, Wilms, neuroblastoma; bowel obstruction or intussusception (ileocaecal Burkitt)
Leg weakness, back pain, incontinence	**Spinal cord compression** (Burkitt, neuroblastoma): an emergency
Fever, night sweats, weight loss (B symptoms)	Lymphoma, leukaemia
Pallor, bruising, nosebleeds, gum bleeding, recurrent infections	Marrow failure: leukaemia, marrow infiltration
Bone pain, limp, refusal to walk, night pain	Leukaemia, neuroblastoma, bone sarcoma

Ask about	Why it matters
White pupil in photos or at night ("cat's eye"), squint, red painful eye, bulging eye	Retinoblastoma
Headache worse in the morning, vomiting, unsteady gait, squint, head enlarging, regression	Brain tumour
Haematuria	Wilms (also schistosomiasis, PSGN: keep a broad view)
HIV status, ART, CD4/viral load; maternal HIV	Kaposi sarcoma, NHL
Family history (retinoblastoma, early cancers)	Heritable syndromes; screen siblings in retinoblastoma
Prior treatment: herbs, steroids, dental extraction, surgery	Steroids partially treat lymphoma or leukaemia and ruin diagnosis; incomplete surgery upstages tumours
Social: distance from Kampala, who can stay, money for transport, who decides	Predicts abandonment; plan support early
Growth, feeding, immunisation (UNEPI), development	Malnutrition worsens treatment toxicity; live vaccines are avoided on chemotherapy

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for children: observe first, examine painful areas gently, and leave distressing parts (mouth, eyes with the ophthalmoscope) to last. In an emergency the ABCDE assessment (section 9) comes first.

Introduction, consent and preparation

- Introduce yourself; confirm the child's name, age and the carer's relationship.
- Explain; consent from the parent and assent from the child.
- **Hand hygiene**.
- Position the child comfortably (**never flat if stridor or orthopnoea**: mediastinal mass); expose in a warm room.

General inspection (end of the bed)

- Well or unwell; respiratory distress (mediastinal mass, effusion); stridor.
- **Pallor, jaundice**, bleeding signs (petechiae, bruises).
- **Wasting**: weight, height, **MUAC**, Z-scores; **body surface area** (needed for chemotherapy and fluids: Mosteller BSA = √(height cm × weight kg ÷ 3600)).
- Obvious masses: jaw, abdomen, eye.
- **Vital signs**: HR, RR, SpO_2, temperature, and **BP** (hypertension in Wilms from renin, and in neuroblastoma from catecholamines).

Hands, face and neck

- **Hands**: pallor, petechiae.
- **Face and mouth**: jaw or maxillary swelling, loose teeth, proptosis, **palatal lesions** (KS: purple), gum hypertrophy (AML).
- **Eyes**: **red reflex** in both eyes with an ophthalmoscope in a dim room from arm's length (absent or white = leukocoria), squint, proptosis, periorbital bruising ("**raccoon eyes**", neuroblastoma).
- **Lymph nodes** (cervical, axillary, inguinal): size and consistency: **rubbery** in Hodgkin, **woody/hard and matted** in KS, **firm and non-tender** in malignancy.

Abdomen

- **Inspection**: distension, visible mass.

- **Palpation**: liver and spleen size; **masses**: site, size (measure with a tape), shape, surface (smooth versus irregular), consistency, tenderness, mobility, moves with respiration, **whether it crosses the midline**, ballotable (renal); check the **other kidney**. **Palpate a suspected Wilms tumour gently and only once** (rupture upstages it).
- **Percussion**: ascites (shifting dullness).
- **Testes**: enlargement (leukaemia, germ cell tumour).

Completing the examination

- **Neurological**: lower-limb power, sensory level, reflexes, anal tone, bladder (cord compression); cranial nerves, papilloedema (brain tumour, CNS Burkitt).
- **Skin**: purple plaques or nodules (KS), "blueberry muffin" nodules (neuroblastoma in infants, leukaemia), petechiae.
- **Bones**: tenderness, swelling, limp.
- Thank, re-dress, **hand hygiene**.
- **Document clearly**: "Right flank mass 10 × 8 cm, smooth, firm, ballotable, does not cross the midline; BP 128/84 (>95th centile)."

Red flags

RED FLAG

- **Spinal cord compression**: back pain, leg weakness, urinary retention. Needs same-day referral (urgent steroids/chemotherapy are decided by the oncologist after diagnosis is secured where possible).
- **Superior mediastinal syndrome**: stridor, orthopnoea, facial swelling, distended neck veins (T-cell ALL/lymphoma). **Do not lay the child flat or sedate them.**
- **Hyperleukocytosis** (WBC >100 × 10^9/L): risk of stroke, respiratory failure and TLS. **Avoid transfusing red cells quickly** (raises viscosity) unless essential.
- **Tumour lysis**: oliguria, high K^+, arrhythmia, seizures, tetany.
- **Febrile neutropenia**: give antibiotics within 60 minutes.
- **Bleeding with platelets <20 × 10^9/L**, DIC (APML).
- **Bowel obstruction / intussusception** in a child older than 3 years: think ileocaecal Burkitt.

Thinking like a clinician

- **Abdominal mass in a well child, smooth, flank, does not cross the midline** → Wilms. **Irregular, crosses the midline, child unwell, bruised eyes, bone pain** → neuroblastoma. **Rapidly growing, multiple masses, ascites, kidneys/ovaries involved, age 5–10** → abdominal Burkitt. **Massive spleen, Karamoja, long fever** → visceral leishmaniasis, not cancer (but rule out leukaemia).
- **Jaw swelling in a 6-year-old** is Burkitt until proven otherwise. Do not extract the teeth.
- **"Malaria not responding"**: pallor, fever, big spleen and liver, with low platelets and a negative RDT → do an FBC and film for **blasts**.
- **Lymphadenopathy**: TB is far more common, but **KS** (HIV, woody nodes, cytopenias), **Hodgkin** (painless rubbery cervical nodes, older child) and **Burkitt** are in the differential. FNA or biopsy before giving TB treatment "trials" if possible.
- **Leukocoria** is retinoblastoma until an ophthalmologist says otherwise. Other causes include congenital cataract, Coats disease, toxocariasis, retinopathy of prematurity.
- **Steroids before diagnosis** are a common, harmful mistake: they shrink lymphoma or leukaemia, hide the diagnosis, and can trigger TLS.

- **Fever in a child on chemotherapy** is febrile neutropenia until the FBC proves otherwise: do not wait for the result before antibiotics.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
Jaw / facial swelling: Burkitt, rhabdomyosarcoma versus dental abscess, osteomyelitis, fibrous dysplasia	Burkitt: rapid growth over days to weeks, loose teeth, not relieved by extraction	FNA/biopsy, OPG/X-ray, LDH
Abdominal mass: Wilms, neuroblastoma, Burkitt, hepatoblastoma versus hydronephrosis, splenomegaly (malaria, leishmaniasis), TB abdomen, full bladder, faecal mass	Wilms smooth, unilateral, well child; neuroblastoma irregular, crosses the midline, unwell; Burkitt multiple masses and ascites	Ultrasound, urine VMA/HVA, FNA
Lymphadenopathy: Hodgkin, NHL/Burkitt, KS, leukaemia versus **TB**, HIV lymphadenopathy, reactive nodes	Rubbery (Hodgkin), woody and matted (KS), cytopenias	FNA/excision biopsy, GeneXpert, HIV test
Pallor, fever, bleeding: leukaemia versus malaria, sepsis, aplastic anaemia, ITP, kala-azar	Hepatosplenomegaly, nodes, bone pain; negative RDT	FBC + film (blasts), marrow
White pupil / squint: retinoblastoma versus cataract, Coats disease, toxocariasis, ROP	Leukocoria under 5 years, family history	Dilated fundoscopy, ultrasound/CT orbit
Bone pain / limp: leukaemia, neuroblastoma, osteosarcoma/Ewing versus osteomyelitis, SCD crisis, rheumatic fever, trauma	Night pain, cytopenias, mass	FBC/film, X-ray, ESR
Skin/palate purple lesions: Kaposi sarcoma versus bacillary angiomatosis, haemangioma, bruising	HIV, woody oedema, nodes	Biopsy, HIV test
Headache, vomiting, ataxia: brain tumour versus meningitis, TB meningitis, hydrocephalus, cerebral malaria	Morning headache, progressive over weeks, focal signs	CT/MRI
Massive splenomegaly: leukaemia versus visceral leishmaniasis (Karamoja), hyper-reactive malarial splenomegaly	Long fever, endemic area	FBC/film; rK39 or splenic aspirate for leishmaniasis

8. Clinical Workup

Tissue is the diagnosis: diagnosis is never made on imaging alone (except Wilms under SIOP and retinoblastoma). "Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes, not an MoH list*.

Bedside

Test	Why	Expected finding	Available at
HemoCue Hb, malaria RDT, glucose	Pallor, fever, coexisting malaria	Low Hb; RDT may be positive (coexisting)	All levels
Urine dipstick	Wilms	Haematuria, protein	All levels
Red reflex (ophthalmoscope)	Leukocoria	Absent or white reflex	All levels with an ophthalmoscope

Test	Why	Expected finding	Available at
HIV rapid test	KS, NHL; ART	Positive	All levels
Point-of-care ultrasound	Mass origin (kidney, adrenal, liver), ascites, IVC	Organ of origin	Where a trained operator is available
Bedside clotting test	DIC (APML, sepsis)	Prolonged clotting	All levels

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
FBC + peripheral film	Every suspected cancer	Blasts (leukaemia), pancytopenia, anaemia, high WBC	Leukaemia diagnosis; transfusion and platelet needs	HC IV and hospitals (film needs an experienced reader)
LDH, uric acid, U&E, creatinine, K^+, phosphate, Ca^{2+}	Burkitt, leukaemia, any large tumour	High LDH (tumour burden), TLS markers	TLS prevention and grading	Hospitals/RRH
HIV test, CD4, viral load	KS, NHL, every child with cancer	Positive; low CD4	ART, cotrimoxazole prophylaxis	HIV test at all levels; CD4/viral load via hub system
G6PD (if rasburicase planned)	High TLS risk	Deficient	Rasburicase contraindicated	Referral labs

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Abdominal ultrasound	Any abdominal mass	Renal origin with claw sign (Wilms), adrenal mass with calcification (neuroblastoma), multiple masses and ascites (Burkitt), IVC thrombus	First-line imaging; distinguishes Wilms, neuroblastoma, Burkitt	Most hospitals and many HC IVs
Chest X-ray	Wilms, lymphoma, leukaemia, osteosarcoma	Lung metastases, mediastinal mass, effusion	Staging; anaesthetic and sedation risk	Hospitals
X-ray / OPG of the jaw	Jaw swelling	Bone destruction, displaced teeth	Supports Burkitt	Hospitals
CT / MRI	Staging Wilms, neuroblastoma, CNS tumours, orbit	Extent, nodes, metastases	Treatment planning	UCI/Mulago /private

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Bone marrow aspirate ± trephine	Suspected leukaemia; Burkitt and neuroblastoma staging	>20% blasts (WHO) / ≥25% lymphoblasts in ALL protocols; tumour infiltration	Confirms and stages	UCI/Mulago, some RRHs
Flow cytometry	Leukaemia typing	B-ALL versus T-ALL versus AML	Protocol choice	UCI/Mulago/private labs

Test	When	Expected / abnormal finding	How it changes management	Available at
FNA cytology or core/excision biopsy	Lymph nodes, jaw/abdominal masses	Starry sky (Burkitt), Reed–Sternberg cells (Hodgkin), spindle cells (KS)	Diagnosis. **Do NOT biopsy a suspected Wilms tumour or retinoblastoma** before oncology/ophthalmology review	RRH (FNA), UCI/Mulago
CSF (lumbar puncture) cytology	Burkitt, ALL (staging)	Malignant cells	CNS-directed therapy. Only after platelets are adequate and no mass effect	UCI, RRH
Urine catecholamines (VMA, HVA)	Suspected neuroblastoma	Raised in >90%	Diagnosis	Referral labs
Hepatitis B, echocardiography	Before chemotherapy (anthracyclines)	Baseline	Doxorubicin cardiotoxicity monitoring	UCI
Examination under anaesthesia with dilated fundoscopy	Suspected retinoblastoma	Intraocular tumour, extent	Enucleation versus chemotherapy	Mulago Eye

Putting it together

Model one-liner: "A 6-year-old boy, 17 kg (WAZ −1.8), with a 3-week history of a rapidly enlarging right maxillary swelling and loose molars after an extraction, with abdominal distension and ascites, LDH 1,450 U/L and uric acid 520 µmol/L: **probable endemic Burkitt lymphoma, likely Murphy stage III, at high risk of tumour lysis**."

Problem list

1. Suspected Burkitt lymphoma needing tissue diagnosis and staging.
2. High TLS risk (bulky disease, high LDH, hyperuricaemia).
3. Moderate malnutrition.
4. Family lives 300 km from UCI (abandonment risk).

Model case presentation (ward round)

> "**Okello is a 7-year-old boy from Arua weighing 19 kg** (WAZ −1.7, HAZ −0.9, BMI-for-age Z −1.9, MUAC 15.2 cm). He is **immunised per UNEPI**, eats 2 meals a day of posho and beans, and is **HIV-negative**. He presents with a **4-week history of a painless left mandibular swelling**, loose lower molars after a dental extraction failed to help, and **2 weeks of abdominal swelling**. On examination he is pale, with a 6 × 5 cm firm non-tender left jaw mass, a 12 cm irregular abdominal mass with shifting dullness, no spinal tenderness and normal lower-limb power. Hb 9.1 g/dL, WBC 7.2, platelets 260, **LDH 1,850 U/L, uric acid 498 µmol/L, K^+ 4.6, creatinine 58 µmol/L**. My working diagnosis is **endemic Burkitt lymphoma, probable stage III**, with **laboratory evidence of high TLS risk**. I have started **hyperhydration at 3 L/m²/day without potassium** and **allopurinol 10 mg/kg/day in 3 doses**, requested an FNA of the jaw and ascitic cytology, and arranged urgent referral to UCI, with social work for transport support."

Worked example 1: staging Burkitt. Jaw mass only → stage I. Jaw mass plus extensive abdominal disease and ascites → stage III. Malignant cells in CSF or marrow → stage IV.

Worked example 2: is this TLS? Day 2 of chemotherapy: uric acid 510 µmol/L, K^+ 6.2 mmol/L, phosphate 2.4 mmol/L, calcium 1.9 mmol/L, creatinine 1.6 × upper limit of normal. Three laboratory criteria (urate, K^+, phosphate) = **laboratory TLS**; creatinine ≥1.5 × ULN makes it **clinical TLS**. Act: stop all K^+, ECG, hyperkalaemia treatment, maximise hydration if not overloaded, discuss rasburicase (if G6PD normal) and dialysis.

Worked example 3: hyperhydration volume. A 19 kg, 115 cm child: BSA = √(115 × 19 ÷ 3600) ≈ 0.78 m². At 3 L/m²/day → about 2.3 L/day (≈ 97 mL/h) without potassium. Target urine output 80–100 mL/m²/h ≈ 62–78 mL/h.

9. Management

Immediate / emergency

First 5 minutes (ABCDE)

1. **A and B**: stridor or orthopnoea (mediastinal mass) → sit up, oxygen, **avoid sedation and lying flat**, call senior.
2. **C**: shock (sepsis, bleeding, tumour rupture) → fluid bolus 10–20 mL/kg (not in SAM; caution with severe anaemia, where blood is the answer).
3. **Fever in a known oncology patient**: assume **febrile neutropenia** → cultures and IV antibiotics within 60 minutes.
4. **D**: check **glucose**, **Hb (HemoCue)**, malaria RDT; **neurology**: leg power, sensory level, bladder (cord compression).
5. **E and urine**: urine output (catheterise or weigh nappies if oliguric: TLS, obstruction).
6. Start **IV access and hydration** if a large tumour is suspected (TLS risk) and give **analgesia**.

Treating established TLS

Problem	Treatment
Hyperkalaemia (K^+ ≥6.0 or ECG changes)	Stop all K^+. **Calcium gluconate 10% 0.5 mL/kg IV over 5–10 min (max 20 mL)** with ECG monitoring if ECG changes; **salbutamol nebulised 2.5 mg (<25 kg) or 5 mg (≥25 kg)**; **insulin 0.1 unit/kg with glucose 0.5 g/kg** (5 mL/kg of 10% dextrose) IV over 30 min, check glucose hourly; sodium polystyrene sulfonate/calcium resonium 1 g/kg oral or rectal; **dialysis** if refractory
Hyperphosphataemia	Limit phosphate intake; oral phosphate binder (e.g. calcium carbonate with meals, aluminium hydroxide short term); dialysis if severe
Hypocalcaemia	**Treat only if symptomatic** (tetany, seizures, arrhythmia): calcium gluconate 10% 0.5 mL/kg IV slowly. Giving calcium when phosphate is high risks calcium phosphate precipitation
AKI / oliguria	Fluid balance, avoid nephrotoxins, adjust drug doses; **dialysis** (peritoneal dialysis is the usual option for children in Uganda) for refractory hyperkalaemia, overload, acidosis or rising creatinine

Febrile neutropenia (and fever in any child on chemotherapy)

1. **Triage immediately**; assume neutropenia until the FBC returns.
2. **Blood culture** (peripheral ± central line), FBC, malaria RDT/film, urine, CXR if respiratory signs; do not delay antibiotics for tests.
3. **Empirical IV antibiotics within 60 minutes** (drug table): ceftriaxone + gentamicin (common Ugandan first line) or ceftazidime or piperacillin–tazobactam; add cloxacillin/vancomycin for line or skin infection, metronidazole for abdominal pain, typhlitis or perianal infection.
4. Treat **malaria** if positive (IV artesunate if severe).
5. Persistent fever after 4–7 days of antibiotics with neutropenia → consider **antifungal** (fluconazole/amphotericin B) at specialist level.
6. **Prevention**: hand hygiene, safe food and water, avoid crowded wards, **cotrimoxazole PCP prophylaxis** in ALL (and HIV), mouth care.

Definitive

At first contact (HC III/IV or general hospital)

1. **Stabilise**: ABC, oxygen, treat shock, sepsis, hypoglycaemia, severe anaemia.
2. **Do no harm**:
   - **No steroids** before diagnosis (unless life-threatening airway compression, and then only after discussing with the oncologist).
   - **No dental extractions** for jaw swellings; no incisional biopsy of an eye mass; **no needle biopsy or excision of a suspected Wilms tumour** outside a specialist centre.
   - **Do not repeatedly palpate** an abdominal tumour.
3. **Basic tests** (FBC, RDT, HIV, ultrasound if available).
4. **Start TLS prevention** for suspected Burkitt or leukaemia with a high WBC: **IV or oral hydration and allopurinol** (TLS prevention below).
5. **Treat pain** and **infection**.
6. **Refer urgently** (same week; retinoblastoma within **72 hours** per UCG 2023) to UCI paediatric oncology (or Mulago Eye for retinoblastoma) with a referral note: history, findings, measurements, results, treatment given, HIV status, weight and height.
7. **Counsel** the family: cancer in children is often curable; treatment is free at UCI; finish all the treatment.

Tumour lysis syndrome: prevention

Step	What to give	Dose	Notes
Hyperhydration	0.9% saline or 0.45% saline/5% dextrose **without potassium** (no Ringer's lactate)	**2–3 L/m²/day** (≈ 2× maintenance; in children <10 kg about 200 mL/kg/day) starting **24–48 h before** chemotherapy and continuing 48–72 h after	Target urine output **80–100 mL/m²/h** (≥3–4 mL/kg/h if <10 kg). Monitor weight, BP and input/output **every 6–8 h**; watch for overload
Allopurinol	Oral	**10 mg/kg/day in 3 divided doses** (or 100 mg/m² 8-hourly), **max 800 mg/day**	Start 24–48 h before chemotherapy; continue 3–7 days. Reduce dose in renal impairment and with mercaptopurine/azathioprine. Prevents new uric acid only
Rasburicase	IV over 30 min	**0.2 mg/kg once daily** for up to 5 days	High-risk patients or uric acid already high. **Contraindicated in G6PD deficiency** (haemolysis, methaemoglobinaemia); check G6PD first. Expensive and rarely available in Uganda
Diuretic	Furosemide	0.5–1 mg/kg IV	Only if fluid overloaded and not hypovolaemic
Monitoring	K^+, PO_4, Ca^{2+}, uric acid, creatinine	Every 6–12 h in high risk for the first 72 h	ECG monitoring if K^+ ≥6.0
Urinary alkalinisation	None	**Not recommended**	It reduces calcium phosphate solubility and is no longer advised

Disease-specific outlines (what happens at UCI)

Cancer	Key diagnosis	Treatment in Uganda (outline)	Outlook
Burkitt lymphoma	FNA/biopsy (starry sky), marrow, CSF, LDH	**Cyclophosphamide-based combination** (e.g. **COM**: cyclophosphamide, vincristine [Oncovin], methotrexate) with **intrathecal chemotherapy**; adding **rituximab (R-COM)** improved 1-year event-free survival in East Africa (67% vs 43%, *Blood Advances* 2025). Surgery only for complications (obstruction). TLS prevention is essential	Highly curable when localised; survival in East Africa is lower in advanced stage and with abandonment

Cancer	Key diagnosis	Treatment in Uganda (outline)	Outlook
ALL	FBC/film, marrow, flow cytometry, CSF	Multi-phase chemotherapy over **2–2.5 years** (induction with steroid, vincristine, asparaginase ± anthracycline; consolidation; CNS prophylaxis with intrathecal methotrexate; maintenance with oral mercaptopurine and methotrexate). Cotrimoxazole prophylaxis	~90% survival in high-income countries; lower but improving in Uganda
AML	Marrow, flow; Auer rods; chloroma (green orbital/skin masses)	Intensive chemotherapy (cytarabine, anthracycline); high toxicity	Poorer prognosis
Wilms tumour	Ultrasound (renal origin, IVC), CXR/CT (lungs). **SIOP approach: no biopsy needed for typical cases**	**SIOP-style** (used in Africa, adapted by SIOP-PODC): **pre-operative chemotherapy** (vincristine + actinomycin-D for about 4 weeks in localised disease) → **nephrectomy** → post-operative chemotherapy ± radiotherapy by stage and histology. (The North American COG approach is upfront nephrectomy)	>90% in high-income countries; lower in SSA due to late presentation and abandonment
Neuroblastoma	Ultrasound/CT (adrenal, calcification), urine VMA/HVA, marrow, biopsy	Depends on risk group: observation (some infant tumours regress, stage MS), surgery, chemotherapy, or intensive multimodal therapy for high-risk	Good if <18 months and localised; poor for high-risk metastatic
Retinoblastoma	Dilated fundoscopy under anaesthesia, ultrasound/CT/MRI orbits. **No biopsy** (risk of seeding)	**Enucleation** for advanced unilateral disease; chemotherapy (vincristine, etoposide, carboplatin) for bilateral/extraocular; focal therapy (laser, cryotherapy) for small tumours; **prosthetic eye**	Eye and life saved if intraocular; extraocular disease has poor survival
Kaposi sarcoma	Biopsy; HIV test, CD4, viral load	**ART** for all + chemotherapy (e.g. bleomycin + vincristine, paclitaxel, or doxorubicin) per UCI protocol	Lymphadenopathic and mild skin forms respond well; woody oedema and visceral forms worse
Hodgkin lymphoma	Excision node biopsy (Reed–Sternberg cells)	Combination chemotherapy (e.g. ABVD/COPP-type) ± radiotherapy	>80–90% curable

DRUG DOSING

Never give intrathecal vincristine. Vincristine given intrathecally is uniformly fatal. It must be given **IV only**, ideally in a minibag, and kept physically apart from intrathecal drugs.

Drug table

Drug	Dose (paediatric per kg, max; adult)	Route	Frequency	Duration	Key cautions
Allopurinol	**10 mg/kg/day** in 3 doses (or 100 mg/m² 8-hourly); **max 800 mg/day**	Oral	8-hourly	From 24–48 h before chemotherapy for 3–7 days	Reduce in renal impairment and with mercaptopurine/azathioprine

Drug	Dose (paediatric per kg, max; adult)	Route	Frequency	Duration	Key cautions
Rasburicase	**0.2 mg/kg**	IV over 30 min	Once daily	Up to 5 days	**Contraindicated in G6PD deficiency**; expensive, rarely available
Hyperhydration fluid (0.9% saline or 0.45% saline/5% dextrose, no K^+)	**2–3 L/m²/day** (<10 kg: ~200 mL/kg/day)	IV (or oral)	Continuous	24–48 h before to 48–72 h after chemotherapy	No Ringer's lactate; watch for overload
Furosemide	0.5–1 mg/kg	IV	As needed	During hyperhydration	Only if overloaded and not hypovolaemic
Calcium gluconate 10%	**0.5 mL/kg** (max 20 mL)	IV over 5–10 min	Repeat if ECG changes persist	Acute	ECG monitoring; in hypocalcaemia treat only if symptomatic
Salbutamol (hyperkalaemia)	**2.5 mg (<25 kg); 5 mg (≥25 kg)**	Nebulised	Repeat as needed	Acute	Tachycardia
Insulin + glucose	**Insulin 0.1 unit/kg + glucose 0.5 g/kg** (5 mL/kg of 10% dextrose)	IV over 30 min	As needed	Acute	Check glucose hourly
Calcium resonium / sodium polystyrene sulfonate	**1 g/kg**	Oral or rectal	Divided doses	Until K^+ controlled	Constipation
Ceftriaxone (febrile neutropenia)	**80–100 mg/kg** (max 4 g)	IV	Once daily	Until afebrile and ANC recovering	Follow UCI protocol
Gentamicin (with ceftriaxone)	**7.5 mg/kg**	IV	Once daily	As above	Check renal function; avoid in TLS-related AKI
Ceftazidime	**50 mg/kg** (max 2 g/dose)	IV	8-hourly	As above	Preferred if *Pseudomonas* risk
Piperacillin–tazobactam	**90 mg/kg** (piperacillin) (max 4.5 g/dose)	IV	6-hourly	As above	International first line; limited availability in Uganda
Cloxacillin	**50 mg/kg** (max 2 g)	IV	6-hourly	As above	Line or skin/soft tissue infection; vancomycin if MRSA risk
Metronidazole	**7.5 mg/kg**	IV	8-hourly	As above	Abdominal pain, typhlitis, perianal infection
Ondansetron	**0.15 mg/kg (max 8 mg)**	IV or oral	Before chemotherapy, then 8–12-hourly	During chemotherapy	QT prolongation

Drug	Dose (paediatric per kg, max; adult)	Route	Frequency	Duration	Key cautions
Cotrimoxazole (PCP prophylaxis)	**5 mg/kg/day trimethoprim component** (max 160 mg)	Oral	2–3 days a week or daily per protocol	Throughout ALL therapy (and HIV)	Marrow suppression
Lactulose (vincristine constipation)	**0.5 mL/kg**	Oral	Twice daily	While constipated	With fluids
Paracetamol	**15 mg/kg**; max 60 mg/kg/day (max 4 g)	Oral	6-hourly	As needed	Masks fever: check temperature before giving in neutropenia
Ibuprofen	**5–10 mg/kg**; max 30–40 mg/kg/day (max 2.4 g)	Oral	6–8-hourly	As needed	**Avoid if platelets low, bleeding or renal impairment**
Oral morphine (immediate release)	**1–12 years: 0.2–0.4 mg/kg every 4 h** (UCG range); 1–12 months: 0.08–0.2 mg/kg every 4 h; **no ceiling dose**, titrate	Oral	4-hourly	As long as needed	Add a laxative; monitor sedation
IV morphine	**0.05–0.1 mg/kg** (max 2.5–5 mg)	IV, slowly	Repeat as needed	Acute	Monitor sedation and RR

Adult doses are not given: dosing in this topic is weight- or BSA-based up to the stated maxima, and chemotherapy doses follow UCI protocols.

Avoid: **codeine and tramadol** in children under 12 years (unpredictable metabolism); steroids before diagnosis; urinary alkalinisation; intrathecal vincristine.

Referral criteria

Level	What to do	Refer up when
HC III	Suspect cancer (SILUAN); Hb, RDT, HIV test; red reflex; treat pain and infection; counsel	Any suspected cancer → hospital or directly to RRH/UCI the same week
HC IV / General Hospital	FBC, ultrasound, CXR; start hydration and allopurinol for a big Burkitt or leukaemia; treat sepsis, anaemia, pain	All suspected cancers → RRH or UCI (same week)
RRH	FBC and film, ultrasound, FNA, LDH and electrolytes; Lacor Burkitt programme	Tissue diagnosis, staging and chemotherapy → UCI
NRH: UCI paediatric oncology (Mulago); Mulago Eye for retinoblastoma	Marrow, flow cytometry, CT/MRI, chemotherapy (free), radiotherapy, surgery, palliative care	n/a

Level split adapted from UCG 2023 levels of care Verify *UCG level codes*.

- **Retinoblastoma**: refer every suspected case to an ophthalmologist **within 72 hours** (UCG 2023).
- **Referral note**: history, findings, measurements, results, treatment given, HIV status, weight and height.

Monitoring and follow-up

- **TLS risk**: weight, BP and input/output every 6–8 h; K^+, PO_4, Ca^{2+}, uric acid, creatinine every 6–12 h for the first 72 h in high risk; ECG if K^+ ≥6.0.
- **During chemotherapy**: FBC before each cycle; temperature checks at home with a plan to return immediately for fever; **MUAC/WHZ weekly**.
- **Abandonment prevention**: social work, transport support, a named contact, tracing of missed appointments (missing 4 consecutive weeks = abandonment).
- **Survivorship**: growth, fertility, cardiac function after anthracyclines, second cancers, school.

Supportive care during chemotherapy

Need	Drug / action	Dose
Nausea and vomiting	**Ondansetron**	**0.15 mg/kg (max 8 mg)** IV/oral before chemotherapy, then 8–12-hourly
PCP prophylaxis	**Cotrimoxazole**	**5 mg/kg/day trimethoprim component** (max 160 mg), e.g. on 2–3 days a week or daily per protocol
Anaemia	Packed cells	10 mL/kg if Hb <7 g/dL or symptomatic (caution with hyperleukocytosis)
Thrombocytopenia	Platelets	10–15 mL/kg if <10 × 10^9/L, or <20–50 with bleeding or before procedures
Nutrition	Assess MUAC/WHZ weekly; therapeutic or supplementary feeds; NG feeding	Malnourished children have more toxicity and abandonment
Mouth care	Chlorhexidine mouthwash, nystatin for thrush	Per protocol
Constipation (vincristine)	Lactulose, fluids	0.5 mL/kg twice daily
Vaccines	Avoid **live vaccines** (measles, BCG, OPV, yellow fever) during chemotherapy and for 3–6 months after; give inactivated vaccines per oncologist	n/a

Pain and palliative care

WHO uses a **two-step approach for persisting pain in children** (WHO 2012 guidance on persisting pain in children; UCG 2023 section 13.1.2 follows the same principle of oral morphine for moderate–severe pain):

Step	Drug	Dose	Max
Mild pain	**Paracetamol**	**15 mg/kg** oral every 6 h	60 mg/kg/day (max 4 g)
	Ibuprofen (avoid if platelets low or bleeding, renal impairment)	**5–10 mg/kg** every 6–8 h	30–40 mg/kg/day (max 2.4 g)
Moderate–severe pain	**Oral morphine** (immediate release)	**1–12 years: 0.2–0.4 mg/kg every 4 h** (UCG range); 1–12 months: 0.08–0.2 mg/kg every 4 h; titrate	No ceiling dose; titrate to effect; add a laxative
	IV morphine	0.05–0.1 mg/kg (max 2.5–5 mg) slowly, repeat as needed	Monitor sedation and RR

- Avoid **codeine and tramadol** in children under 12 years (unpredictable metabolism).

- **Palliative care** is part of care from diagnosis, not just at the end: pain, nutrition, psychosocial and spiritual support, and **end-of-life care at home** when cure is not possible.

10. Complications

Disease-related

- Tumour lysis syndrome, AKI.
- Spinal cord compression, raised ICP.
- Airway compression (mediastinal mass).
- Bowel obstruction, intussusception, perforation.
- Hyperleukocytosis, DIC, bleeding.
- Bone marrow failure.
- Blindness, loss of an eye (retinoblastoma).

Treatment-related

Complication	Cause	Recognition and action
Febrile neutropenia, sepsis (Gram-negative, *Staph*, fungi)	Chemotherapy	Antibiotics within 60 min; antifungal if persistent fever
Mucositis, malnutrition	Chemotherapy	Mouth care, feeds, NG feeding
Cardiotoxicity	Doxorubicin	Baseline and follow-up echo
Nephrotoxicity, ototoxicity	Cisplatin, ifosfamide	Renal function, hearing
Neuropathy, constipation, jaw pain	Vincristine	Lactulose; never intrathecal
Hepatotoxicity, pancreatitis	Methotrexate, mercaptopurine; asparaginase	LFTs; abdominal pain
TLS	Chemotherapy in bulky disease	Prevention (section 9)
Haemolysis, methaemoglobinaemia	Rasburicase in G6PD deficiency	Check G6PD first
Infertility, second cancers, growth problems	Chemotherapy, radiotherapy	Survivorship clinic
Psychosocial distress, school disruption, **abandonment**	Long treatment far from home	Social support, counselling

11. Prognosis

- **Over 80%** of children with cancer survive in high-income countries, but **under 30%** in many LMICs, mainly because of **late diagnosis and abandoned treatment**.
- **Burkitt**: highly curable when localised; R-COM improved 1-year event-free survival in East Africa (67% vs 43%); worse in advanced stage and with abandonment.
- **ALL**: ~90% survival in high-income countries; lower but improving in Uganda. **AML**: poorer.
- **Wilms**: >90% in high-income countries; lower in SSA due to late presentation and abandonment.
- **Neuroblastoma**: good if under 18 months and localised; poor for high-risk metastatic disease.
- **Retinoblastoma**: eye and life saved if intraocular; extraocular disease has poor survival.
- **Kaposi sarcoma**: lymphadenopathic and mild skin forms respond well; woody oedema and visceral forms worse.
- **Hodgkin lymphoma**: >80–90% curable.

12. Prevention

- **Early diagnosis is the main "prevention" of death**: use **SILUAN** warning signs, examine the red reflex, and refer the same week.
- **HIV**: early infant diagnosis and ART reduce Kaposi sarcoma and HIV-associated NHL.
- **Malaria control** may reduce the EBV–malaria interaction behind endemic Burkitt.
- **Retinoblastoma**: screen siblings and offspring of children with heritable (bilateral) disease.
- **Prevent treatment deaths**: TLS prophylaxis, prompt febrile neutropenia care, nutrition support.
- **Prevent abandonment**: counsel that treatment is free and often curative, plan transport and food support early.

13. Special Populations

- **HIV-positive children**: Kaposi sarcoma (lymphadenopathic, woody oedema, palate) and NHL; ART for all, cotrimoxazole prophylaxis.
- **Down syndrome**: 10–20× risk of ALL; AML (especially megakaryoblastic).
- **Severe acute malnutrition**: no fluid boluses; more chemotherapy toxicity and abandonment; therapeutic feeding.
- **G6PD deficiency**: rasburicase contraindicated.
- **Infants under 10 kg**: hyperhydration about 200 mL/kg/day; urine target ≥3–4 mL/kg/h; neuroblastoma stage MS may regress; morphine 0.08–0.2 mg/kg every 4 h (1–12 months).
- **Children under 12 years**: avoid codeine and tramadol.
- **Children on chemotherapy**: no live vaccines during and for 3–6 months after.

14. Key Points

High-yield points

- A lump, white pupil, unexplained bruising or pallor, persistent bone pain or a rapidly growing jaw swelling needs a **diagnosis this week**.
- **Jaw swelling in a 6-year-old = Burkitt**: do not extract teeth.
- **Wilms is well and stays on one side; neuroblastoma is nasty and crosses the line.** Palpate Wilms gently and once; no local biopsy.
- **Leukocoria = retinoblastoma**: ophthalmologist within 72 hours.
- **No steroids before diagnosis.**
- **TLS prevention**: 2–3 L/m²/day without K^+, urine 80–100 mL/m²/h, allopurinol 10 mg/kg/day; no urinary alkalinisation.
- **Febrile neutropenia**: IV antibiotics within 60 minutes.
- **Never intrathecal vincristine.**
- Oral morphine has no ceiling dose; avoid codeine and tramadol under 12.
- Treatment at UCI is free; preventing abandonment saves lives.

Numbers & doses

Item	Value
Children with cancer per year worldwide (0–19 y)	~400,000
Survival HIC vs many LMIC	>80% vs <30%

Item	Value
GICC target	≥60% survival by 2030
Uganda: children among cancer patients	~10% (UCG 2023)
Burkitt doubling time	~25 hours
Burkitt peak age (UCG)	4–7 years, boys > girls
Wilms typical age	2–5 years (UCG average 2)
Neuroblastoma median age	~18 months
Retinoblastoma	Usually <3–5 years; refer within 72 h
ALL share of childhood leukaemia	~75–80%; peak 2–5 years
Hyperleukocytosis	WBC >100 × 10^9/L
TLS hyperhydration	2–3 L/m^2/day, no K^+
Target urine output	80–100 mL/m^2/h
Allopurinol	10 mg/kg/day ÷ 3 (max 800 mg/day)
Rasburicase	0.2 mg/kg IV daily; not in G6PD deficiency
TLS lab thresholds	Urate ≥476 µmol/L; K^+ ≥6.0; PO_4 ≥2.1 (child); Ca ≤1.75 mmol/L
Febrile neutropenia	ANC <0.5; T ≥38.3 once or ≥38.0 for 1 h; antibiotics within 60 min
Calcium gluconate 10% (hyperkalaemia)	0.5 mL/kg IV over 5–10 min (max 20 mL)
Ondansetron	0.15 mg/kg (max 8 mg)
Oral morphine (1–12 y)	0.2–0.4 mg/kg every 4 h, titrate
Mosteller BSA	√(height cm × weight kg ÷ 3600)
Abandonment	≥4 consecutive weeks of missed curative therapy

Exam traps

- **Trap**: Extracting loose teeth in a child with a jaw swelling. **Right**: suspect Burkitt; refer for FNA/biopsy.
- **Trap**: Biopsying a suspected Wilms tumour locally. **Right**: SIOP-style care treats typical Wilms with pre-operative chemotherapy without biopsy; biopsy or rupture upstages it.
- **Trap**: Prednisolone "for the swollen nodes" before diagnosis. **Right**: no steroids until a tissue diagnosis.
- **Trap**: Alkalinising the urine in TLS. **Right**: no longer recommended; hyperhydrate and give allopurinol/rasburicase.
- **Trap**: Giving IV calcium to every child with low calcium in TLS. **Right**: only if symptomatic, because of calcium phosphate precipitation.
- **Trap**: Rasburicase in a boy with G6PD deficiency. **Right**: contraindicated (haemolysis, methaemoglobinaemia).
- **Trap**: Treating a white pupil as conjunctivitis. **Right**: leukocoria = retinoblastoma until proven otherwise.
- **Trap**: Waiting for the FBC before antibiotics in a febrile child on chemotherapy. **Right**: antibiotics within 60 minutes.
- **Trap**: Laying a child with stridor and a mediastinal mass flat for a CXR or sedating them. **Right**: keep upright, no sedation.

Questions seniors ask

Viva questions

1. **Why does Burkitt lymphoma respond so well to chemotherapy?** Nearly all cells are dividing (growth fraction ~100%), so cell-cycle drugs kill them efficiently. The same property causes TLS.
2. **How do you tell Wilms from neuroblastoma clinically?** Wilms: well child, smooth flank mass, does not cross the midline, haematuria/hypertension. Neuroblastoma: younger, unwell, irregular mass crossing the midline, bone pain, raccoon eyes, raised VMA/HVA, calcification.
3. **What is leukocoria and what do you do?** A white pupillary reflex instead of red; refer to an ophthalmologist urgently (UCG: within 72 h) for retinoblastoma.
4. **Why is allopurinol not enough for established hyperuricaemia?** It stops new uric acid production but does not remove what is already there; rasburicase breaks down existing uric acid.
5. **Why must we avoid steroids in a child with suspected leukaemia or lymphoma?** They can induce partial remission, obscuring the diagnosis and immunophenotype, and can precipitate TLS.
6. **What is the definition of febrile neutropenia and your first action?** ANC $<0.5 \times 10^9$/L with temperature ≥38.3 °C once or ≥38.0 °C for 1 h; cultures then IV antibiotics within 60 minutes.

On the ward round

1. **What are the three types of Burkitt lymphoma?** Endemic (Africa, EBV, jaw), sporadic (abdomen), immunodeficiency-associated (HIV).
2. **Which cancers are WHO GICC index cancers?** ALL, Burkitt, Hodgkin, retinoblastoma, Wilms, low-grade glioma.
3. **How do you prevent TLS in a child with Burkitt?** Hyperhydration 2–3 L/m²/day without potassium, allopurinol 10 mg/kg/day (or rasburicase if high risk and G6PD normal), monitoring of electrolytes and urine output.
4. **What associations does Wilms tumour have?** WAGR, Denys–Drash, Beckwith–Wiedemann, hemihypertrophy, aniridia.
5. **What are the features of neuroblastoma outside the abdomen?** Periorbital ecchymosis, proptosis, Horner syndrome, opsoclonus–myoclonus, bone pain, blueberry-muffin skin, hypertension, diarrhoea (VIP).
6. **Why do Ugandan children with cancer do worse?** Late presentation, misdiagnosis, abandonment, infection, malnutrition, limited radiotherapy and pathology, distance.
7. **What is your referral pathway?** HC → general hospital → RRH → Uganda Cancer Institute (Mulago Eye for retinoblastoma), with urgent referral and a clear note.

Mnemonics

- **SILUAN early warning signs** (used in childhood cancer awareness campaigns): **S**eek help early for persistent symptoms; **I**nsight: white spot in the eye, new squint, blindness, bulging eye; **L**ump in abdomen, pelvis, head, neck, limbs, testes or glands; **U**nexplained fever (>2 weeks), weight loss, pallor, fatigue, bruising or bleeding; **A**ching bones, joints or back, easy fractures; **N**eurological signs: change in walk, balance or speech, regression, headache >1 week with or without vomiting, enlarging head.
- **TLS = "PUKE Ca down"**: **P**hosphate up, **U**ric acid up, **K**$^+$ up, **E**nd-organ (creatinine, arrhythmia, seizures), **Ca**lcium down.
- **Wilms versus neuroblastoma**: "**W**ilms is **W**ell and stays on one side; **N**euroblastoma is **N**asty and crosses the line."
- **Burkitt: "8-14, the jaw is seen"** (t(8;14), jaw tumour).
- **Retinoblastoma: "13 makes a white eye"** (RB1 on chromosome 13).

OSCE tips

- **Abdominal mass station**: always state **site, size (cm), shape, surface, consistency, tenderness, mobility, moves with respiration?, ballotable?, crosses the midline?** and check **BP** and the **other kidney**.
- **Eye station**: do the **red reflex test** in a dim room from arm's length, comparing both eyes.

- **Counselling station**: explain the diagnosis is suspected, not confirmed; treatment at UCI is free; many children are cured; the importance of completing treatment; ask about support needs.

15. References

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SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Why do far fewer children with cancer survive in LMICs than in high-income countries?	**Late diagnosis/misdiagnosis**, inaccessible therapy, **abandonment of treatment**, death from toxicity (infection, TLS, malnutrition) and relapse. Survival is >80% in HICs vs <30% in many LMICs (WHO).
2	WHO GICC (CureAll) target and its six index cancers?	**≥60% survival by 2030**. Index cancers: **ALL, Burkitt lymphoma, Hodgkin lymphoma, retinoblastoma, Wilms tumour, low-grade glioma**.
3	Burkitt lymphoma: genetics and doubling time?	**MYC** translocation, usually **t(8;14)** (also t(2;8), t(8;22)). Growth fraction ~100%, **doubling time ~25 hours**.
4	Three clinical variants of Burkitt lymphoma?	**Endemic** (equatorial Africa; EBV ~100%; jaw, orbit, abdomen, kidneys, ovaries), **sporadic** (Western; abdominal/ileocaecal), **immunodeficiency-associated** (HIV, transplant).
5	Classic histology of Burkitt lymphoma?	**Starry-sky** pattern (sheets of monotonous B cells with tingible-body macrophages), **Ki-67 ~100%**, CD20+, CD10+, BCL2−.
6	Murphy (St Jude) staging of Burkitt lymphoma in one line each?	**I**: single extranodal/nodal site (not chest/abdomen). **II**: regional nodes or ≥2 sites same side of diaphragm, resectable GI. **III**: both sides of diaphragm, intrathoracic, extensive abdominal, paraspinal/epidural. **IV**: **CNS or bone marrow**.
7	What chemotherapy is used for Burkitt lymphoma in East Africa, and what did the 2025 rituximab study show?	Cyclophosphamide-based **COM** (cyclophosphamide, vincristine, methotrexate) + intrathecal therapy. Adding **rituximab (R-COM)** raised 12-month EFS from **43% to 67%** and reduced deaths (*Blood Advances* 2025).
8	Cairo–Bishop laboratory TLS criteria (children)?	≥2 of: **uric acid ≥476 µmol/L**, **K^+ ≥6.0 mmol/L**, **phosphate ≥2.1 mmol/L**, **Ca^{2+} ≤1.75 mmol/L** (or 25% change from baseline), from 3 days before to 7 days after treatment.
9	Clinical TLS = laboratory TLS plus what?	Any of: **creatinine ≥1.5 × ULN**, **cardiac arrhythmia/sudden death**, or **seizure**.
10	TLS prevention: fluids and urine output target?	**2–3 L/m²/day** isotonic fluid **without potassium** (not Ringer's), starting 24–48 h before chemo; target urine output **80–100 mL/m²/h**.
11	Allopurinol dose for TLS prevention in children?	**10 mg/kg/day in 3 divided doses** (or 100 mg/m² 8-hourly), **max 800 mg/day**, starting 24–48 h before chemotherapy.
12	Rasburicase: dose and key contraindication?	**0.2 mg/kg IV once daily** (up to 5 days). **Contraindicated in G6PD deficiency** (haemolysis, methaemoglobinaemia).
13	Why is urinary alkalinisation no longer recommended in TLS?	It promotes **calcium phosphate** (and xanthine) precipitation in the kidney; hyperhydration with allopurinol/rasburicase is preferred.
14	Define febrile neutropenia.	**ANC <0.5 × 10^9/L** (or expected to fall below) with temperature **≥38.3 °C once** or **≥38.0 °C sustained over 1 hour**. Give IV broad-spectrum antibiotics **within 60 minutes**.
15	Wilms tumour: typical presentation?	Well child aged **2–5 years** with a **smooth, firm flank mass that does not cross the midline**; haematuria and hypertension in ~25%.
16	Syndromes associated with Wilms tumour?	**WAGR** (WT1 deletion 11p13), **Denys–Drash**, **Beckwith–Wiedemann**, isolated **hemihypertrophy**, aniridia.

#	Question	Answer
17	SIOP vs COG approach to Wilms tumour?	**SIOP** (used in Africa/Europe): **pre-operative chemotherapy** (vincristine + actinomycin-D) **without biopsy** for typical cases, then nephrectomy. **COG** (North America): **upfront nephrectomy**.
18	Neuroblastoma: clinical clues?	Age <5 (median ~18 months), **irregular hard mass crossing the midline**, sick child, **periorbital bruising (raccoon eyes)**, proptosis, bone pain, Horner syndrome, opsoclonus–myoclonus, hypertension, blueberry-muffin skin. **Raised urinary VMA/HVA**, calcification on imaging.
19	Retinoblastoma: gene, presentation and UCG action?	**RB1 on 13q14** (two-hit). **Leukocoria**, squint, red eye, proptosis (late). Bilateral = heritable. UCG 2023: **refer urgently within 72 hours** to an ophthalmologist; midwives check eyes at birth.
20	Most common childhood leukaemia and its peak age?	**ALL** (~75–80% of childhood leukaemia), peak **2–5 years**. Down syndrome increases risk 10–20×.
21	Why avoid steroids before diagnosing lymphoma/leukaemia?	They cause partial remission that **masks the diagnosis** (histology/immunophenotype) and can **precipitate TLS**.
22	Paediatric Kaposi sarcoma in Uganda: features and treatment?	HHV-8, usually with HIV. Children often have **lymphadenopathic disease, woody oedema, palatal lesions**, cytopenias, more than classic skin plaques. **ART + chemotherapy** (e.g. bleomycin/vincristine, paclitaxel) at UCI.
23	SILUAN early warning signs of childhood cancer?	**S**eek help early; **I**nsight (white eye, squint, proptosis); **L**ump; **U**nexplained fever, weight loss, pallor, bruising; **A**ching bones/joints/back; **N**eurological signs (gait, balance, speech, headache, enlarging head).
24	Emergency treatment of hyperkalaemia in TLS (first three steps)?	1. **Calcium gluconate 10% 0.5 mL/kg IV over 5–10 min (max 20 mL)** if ECG changes; 2) **salbutamol neb 2.5 mg (<25 kg) / 5 mg**; 3) **insulin 0.1 U/kg + glucose 0.5 g/kg**. Stop K^+; resonium; dialysis if refractory.
25	Which drug must never be given intrathecally?	**Vincristine** (fatal ascending neurotoxicity). IV only, ideally in a minibag, separated from intrathecal drugs.
26	Oral morphine starting dose for cancer pain in a child aged 1–12 years?	**0.2–0.4 mg/kg every 4 hours**, titrated (no ceiling), with a laxative. Avoid codeine and tramadol under 12 years.
27	Who was Denis Burkitt and why does Uganda matter in this story?	An Irish surgeon at **Mulago, Kampala**, who in 1958 described fast-growing jaw tumours in children across the malaria belt; Epstein later found **EBV** in his samples, the first virus linked to a human cancer.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 6-year-old boy (18 kg) from Gulu has a 3-week history of a rapidly enlarging, painless right maxillary swelling. A dental extraction at a clinic did not help and two more molars are loose. He is afebrile and otherwise well. What is the MOST appropriate next step?

A Extract the remaining loose teeth and give amoxicillin

B Arrange FNA/biopsy of the mass, check FBC, LDH, uric acid and HIV, and refer urgently to the cancer centre

C Start prednisolone 2 mg/kg to shrink the swelling before referral

D Treat as a dental abscess with 2 weeks of metronidazole and review

E Start anti-TB treatment because jaw TB is common

2. A 3-year-old girl (13 kg) is noticed by her mother to have a swelling in the left side of her abdomen while bathing. She is playful and eating well. There is a smooth, firm, 9 × 7 cm left flank mass that is ballotable and does not cross the midline. BP is 124/80 mmHg. Urine dipstick shows blood 2+. What is the MOST likely diagnosis?

A Neuroblastoma

B Splenomegaly from malaria

C Abdominal Burkitt lymphoma

D Wilms tumour (nephroblastoma)

E Hydronephrosis from pelviureteric junction obstruction

3. A 7-year-old boy (20 kg, BSA 0.8 m^2) with biopsy-proven stage III Burkitt lymphoma is due to start chemotherapy tomorrow. LDH 2,100 U/L, uric acid 510 µmol/L, K^+ 4.4 mmol/L, creatinine 55 µmol/L. G6PD status unknown and rasburicase is unavailable. Which plan is BEST?

A IV 0.9% saline without potassium at about 2–3 L/m^2/day (≈2 L/day), allopurinol 10 mg/kg/day in 3 doses (≈70 mg 8-hourly), strict input/output and electrolytes every 6–12 h

B Ringer's lactate at maintenance rate and start chemotherapy today

C Sodium bicarbonate infusion to alkalinise the urine to pH >7.5

D Furosemide 2 mg/kg twice daily to keep urine flowing, with fluid restriction

E Give rasburicase from a private pharmacy without checking G6PD

4. A 5-year-old girl (16 kg) receiving induction chemotherapy for ALL at UCI develops a temperature of 38.6 °C. She is alert, HR 128, CRT 2 s. Her last ANC (2 days ago) was 0.3 × 10^9/L. What should happen NEXT?

A Give paracetamol and repeat the temperature in 4 hours

B Wait for today's FBC before deciding about antibiotics

C Give oral amoxicillin–clavulanate and observe

D Do a malaria RDT and treat only if positive

E Take blood cultures and give broad-spectrum IV antibiotics within 60 minutes (e.g. ceftriaxone + gentamicin or ceftazidime per protocol)

5. A 2-year-old boy is brought to a HC IV because his mother noticed his left eye 'shines white like a cat's eye' in photographs for 2 months. He has a slight inward squint. The eye is not red. What is the MOST appropriate action?

A Prescribe chloramphenicol eye ointment and review in 2 weeks
B Reassure: this is a normal reflection in photographs
C Refer urgently to an ophthalmologist (within 72 hours) for suspected retinoblastoma
D Patch the right eye for amblyopia and review in 3 months
E Arrange a fine-needle aspirate of the eye at the regional hospital

6. An 18-month-old girl (8.5 kg) is irritable, has had fever and weight loss, and has bruising around both eyes without trauma. There is a hard, irregular abdominal mass crossing the midline. BP 118/76. Ultrasound shows a suprarenal mass with calcification. Which test BEST supports the diagnosis?

A Urinary catecholamine metabolites (VMA and HVA)
B Serum alpha-fetoprotein
C Malaria blood film
D Hb electrophoresis
E Anti-streptolysin O titre

7. A 4-year-old boy (15 kg) has had 3 weeks of fever, pallor, bone pain and a limp. He has petechiae, cervical lymphadenopathy and the spleen is 4 cm below the costal margin. RDT negative. Hb 6.1 g/dL, WBC 48 × 10^9/L, platelets 22 × 10^9/L. A clinician suggests starting prednisolone because 'it might be rheumatic fever'. What is the BEST response?

A Agree and start prednisolone 2 mg/kg
B Start artemether–lumefantrine because RDTs miss malaria
C Avoid steroids; request a peripheral film for blasts, start TLS precautions, treat infection if febrile, and refer urgently for bone marrow and flow cytometry
D Transfuse whole blood 30 mL/kg rapidly
E Give aspirin 100 mg/kg/day for possible rheumatic fever

8. Twenty-four hours after starting chemotherapy for Burkitt lymphoma, a 9-year-old (25 kg) has K^+ 6.8 mmol/L with peaked T waves, phosphate 2.9 mmol/L, Ca^{2+} 1.6 mmol/L and urine output 0.4 mL/kg/h. What is the FIRST drug to give? Advanced

A Furosemide 1 mg/kg IV
B Sodium bicarbonate 2 mmol/kg IV
C Oral calcium resonium 1 g/kg
D Allopurinol 10 mg/kg
E Calcium gluconate 10% 0.5 mL/kg IV over 5–10 minutes with ECG monitoring

9. A 6-year-old HIV-positive girl on ART for 3 months has firm, matted, 'woody' bilateral cervical and inguinal lymph nodes, swollen legs and a purple lesion on the hard palate. Hb 7.9, platelets 85 × 10^9/L. GeneXpert on sputum is negative. What is the MOST likely diagnosis?

A Tuberculous lymphadenitis
B Kaposi sarcoma
C Hodgkin lymphoma
D Reactive HIV lymphadenopathy
E Filarial lymphoedema

10. A 3-year-old with a suspected Wilms tumour is at a regional referral hospital. Ultrasound shows a 10 cm solid left renal mass and a normal right kidney. The surgeon offers to remove it today because 'the cancer centre is far'. What is the MOST appropriate advice? Advanced

A Proceed with nephrectomy today to avoid delay

B Perform a percutaneous needle biopsy first, then decide

C Start radiotherapy at the regional hospital

D Refer urgently to UCI for staging (chest imaging) and pre-operative chemotherapy per the SIOP-adapted approach, with surgery afterwards

E Watch and wait with a repeat ultrasound in 3 months

11. A 10-year-old boy (28 kg) with a large mediastinal mass on CXR (suspected T-cell lymphoma) has orthopnoea, facial swelling and distended neck veins. A CT scan under sedation is requested. What is the MOST important safety point? Advanced

A Avoid general anaesthesia and deep sedation, keep him sitting up, and obtain diagnosis by the least invasive route (e.g. pleural fluid, peripheral node, marrow)

B Lie him flat with ketamine sedation so the scan is clear

C Give a 20 mL/kg fluid bolus before the scan

D Start prednisolone immediately without any tissue sample in all cases

E Send him home with salbutamol and review the CT result in a week

12. A 5-year-old with metastatic retinoblastoma is receiving palliative care at home in Mbarara. He weighs 16 kg and has severe bone pain despite paracetamol 15 mg/kg every 6 hours. What is the MOST appropriate next analgesic step?

A Add codeine 1 mg/kg every 6 hours

B Add tramadol 1 mg/kg every 6 hours

C Start oral morphine about 3–6 mg (0.2–0.4 mg/kg) every 4 hours, titrated to effect, with a laxative

D Increase paracetamol to 30 mg/kg every 6 hours

E Give pethidine IM when needed

13. A 7-year-old boy with abdominal Burkitt lymphoma develops back pain and, over 12 hours, weakness of both legs and urinary retention. Power is 2/5 in both legs with a sensory level at T10. What is the BEST immediate action?

A Physiotherapy and review in a week

B Lumbar puncture for intrathecal chemotherapy before anything else

C Analgesia only, as this is likely bone pain

D Oral ibuprofen and catheterise

E Treat as malignant spinal cord compression: catheterise, urgent senior/oncology discussion the same day for definitive treatment (chemotherapy ± steroids or surgery per oncologist), and urgent imaging if available

14. A mother from Kasese has stopped bringing her 6-year-old son for ALL maintenance chemotherapy for 6 weeks. She says transport costs and her other children at home made it impossible. Which response BEST reflects good practice?

A Record 'lost to follow-up' and close the file

B Recognise this as treatment abandonment, contact the family, involve social work and support organisations for transport/lodging, re-counsel on curability, and plan to resume treatment

C Tell the mother that stopping means the child will certainly die

D Switch the child to herbal treatment closer to home

E Restart induction chemotherapy from the beginning without discussion

15. Which statement about endemic Burkitt lymphoma is CORRECT?

A It is a T-cell lymphoma with a BCL2 translocation

B It is rarely associated with Epstein–Barr virus

C It most often presents in adolescents aged 15–19 years

D It is a B-cell lymphoma with a MYC translocation, nearly always EBV-positive, and frequently involves the jaw in young children

E It grows slowly over many months, so tumour lysis is rare

Answers & explanations

1. B · Arrange FNA/biopsy of the mass, check FBC, LDH, uric acid and HIV, and refer urgently to the cancer centre

B is correct: a rapidly growing jaw mass with loose teeth in a 6-year-old in the malaria belt is **endemic Burkitt lymphoma** until proven otherwise; get tissue, TLS bloods and HIV status, and refer urgently. A is wrong: extractions do not help (UCG) and delay diagnosis. C is wrong: steroids before diagnosis obscure histology and can trigger tumour lysis. D is wrong: an afebrile, painless, rapidly enlarging mass is not an abscess. E is wrong: empirical TB treatment without tissue diagnosis delays cancer care.

2. D · Wilms tumour (nephroblastoma)

D is correct: a **well** pre-school child with a **smooth, ballotable flank mass not crossing the midline**, hypertension and haematuria is the classic Wilms picture. A is wrong: neuroblastoma usually affects a sick child with an irregular mass that crosses the midline. B is wrong: the spleen has a notch, moves down with respiration and is not ballotable. C is wrong: abdominal Burkitt is usually multifocal with ascites in an older child and grows over days. E is wrong: hydronephrosis can be ballotable but does not usually cause hypertension with haematuria, and ultrasound would show a dilated pelvis rather than a solid mass.

3. A · IV 0.9% saline without potassium at about 2–3 L/m²/day (≈2 L/day), allopurinol 10 mg/kg/day in 3 doses (≈70 mg 8-hourly), strict input/output and electrolytes every 6–12 h

A is correct: he is high risk for TLS (bulky Burkitt, high LDH and uric acid), so he needs **hyperhydration 2–3 L/m²/day without potassium**, allopurinol 10 mg/kg/day divided 8-hourly, and close monitoring before chemotherapy. B is wrong: Ringer's contains potassium and maintenance volume is too little. C is wrong: urinary alkalinisation is no longer recommended (calcium phosphate precipitation). D is wrong: fluid restriction plus diuretic causes dehydration and worsens crystal nephropathy. E is wrong: rasburicase is contraindicated in G6PD deficiency, which is common in Uganda, so it must be checked first.

4. E · Take blood cultures and give broad-spectrum IV antibiotics within 60 minutes (e.g. ceftriaxone + gentamicin or ceftazidime per protocol)

E is correct: this is **febrile neutropenia**, which can progress to septic shock within hours; give IV broad-spectrum antibiotics within 60 minutes after cultures. A is wrong: antipyretics alone delay treatment. B is wrong: assume neutropenia; do not wait for the count. C is wrong: oral antibiotics are not adequate initial therapy for a neutropenic child on induction chemotherapy. D is wrong: test for malaria, but a negative or positive RDT does not remove the need for antibacterial cover.

5. C · Refer urgently to an ophthalmologist (within 72 hours) for suspected retinoblastoma

C is correct: **leukocoria with a squint** in a toddler is retinoblastoma until proven otherwise; UCG 2023 says refer within 72 hours. A is wrong: this is not conjunctivitis. B is wrong: a persistent white reflex is never normal. D is wrong: patching treats amblyopia but misses a life-threatening tumour. E is wrong: biopsy or FNA of a retinoblastoma risks spreading tumour outside the eye.

6. A · Urinary catecholamine metabolites (VMA and HVA)

A is correct: a sick infant with periorbital bruising (raccoon eyes, orbital metastases) and a calcified adrenal mass crossing the midline suggests **neuroblastoma**; urinary VMA/HVA are raised in over 90%. B is wrong: AFP supports hepatoblastoma or germ cell tumours. C is wrong: malaria does not cause a calcified suprarenal mass. D is wrong: electrophoresis diagnoses haemoglobinopathies. E is wrong: ASOT supports post-streptococcal disease, not a tumour.

7. C · Avoid steroids; request a peripheral film for blasts, start TLS precautions, treat infection if febrile, and refer urgently for bone marrow and flow cytometry

C is correct: pancytopenia (low Hb and platelets) with high WBC, organomegaly, nodes and bone pain is **acute leukaemia** until proven otherwise. Steroids before diagnosis mask it and can trigger TLS, so get a film, start precautions and refer. A is wrong for those reasons. B is wrong: the RDT is negative and this picture is not malaria. D is wrong: rapid large-volume blood risks overload and hyperviscosity if the WBC is high; give packed cells cautiously if needed. E is wrong: aspirin is dangerous with platelets of 22 and rheumatic fever does not explain cytopenias and organomegaly.

8. E · Calcium gluconate 10% 0.5 mL/kg IV over 5–10 minutes with ECG monitoring

E is correct: hyperkalaemia with **ECG changes** needs **IV calcium first** to stabilise the myocardium (12.5 mL here, within the 20 mL max), then salbutamol and insulin–glucose to shift potassium, and urgent dialysis planning for oliguric TLS. A is wrong: furosemide does not act fast enough and he is oliguric. B is wrong: bicarbonate is not first line and may worsen hypocalcaemia and calcium phosphate deposition. C is wrong: resins take hours. D is wrong: allopurinol prevents uric acid formation but does nothing for potassium.

9. B · Kaposi sarcoma

B is correct: in Ugandan children with HIV, **Kaposi sarcoma** often presents with **lymphadenopathic disease, woody oedema, palatal purple lesions and cytopenias**; confirm by biopsy. A is wrong: TB nodes are usually caseating and may discharge, and they do not explain the palatal lesion. C is wrong: Hodgkin nodes are rubbery and usually cervical in an older child, without palatal lesions. D is wrong: reactive nodes are small and soft. E is wrong: filariasis causes lymphoedema but not palatal purple lesions or cytopenias.

10. D · Refer urgently to UCI for staging (chest imaging) and pre-operative chemotherapy per the SIOP-adapted approach, with surgery afterwards

D is correct: in African (SIOP-adapted) practice, typical Wilms tumours get **pre-operative chemotherapy** (vincristine + actinomycin-D) at a cancer centre, which shrinks the tumour and reduces rupture; surgery follows in a specialist unit with post-operative chemotherapy by stage. A is wrong: non-specialist upfront nephrectomy risks rupture and spillage, which upstages the tumour and worsens outcome. B is wrong: needle biopsy is not needed for typical cases under SIOP and can upstage. C is wrong: radiotherapy is only at UCI and is given by stage after surgery. E is wrong: delay allows a fast tumour to spread.

11. A · Avoid general anaesthesia and deep sedation, keep him sitting up, and obtain diagnosis by the least invasive route (e.g. pleural fluid, peripheral node, marrow)

A is correct: **superior mediastinal syndrome** means sedation, anaesthesia or lying flat can cause complete airway or cardiovascular collapse; keep him upright and get tissue the least invasive way, with oncology and anaesthesia involved. B is wrong: supine sedation is the classic fatal error. C is wrong: fluid does not relieve compression and can worsen venous congestion. D is wrong: steroids may be needed if life-threatening, but only as a senior/oncology decision, ideally after obtaining a sample, because they destroy the diagnosis. E is wrong: this is an emergency.

12. C · Start oral morphine about 3–6 mg (0.2–0.4 mg/kg) every 4 hours, titrated to effect, with a laxative

C is correct: WHO's two-step approach for children moves straight from non-opioids to a **strong opioid**: oral morphine 0.2–0.4 mg/kg every 4 hours, titrated with no ceiling, plus a laxative. Oral morphine is available free in Uganda for palliative care. A is wrong: codeine is not recommended in children (variable CYP2D6 metabolism). B is wrong: tramadol is avoided under 12 years. D is wrong: 120 mg/kg/day of paracetamol is hepatotoxic. E is wrong: pethidine has a toxic metabolite and IM injections are painful and unsuitable for ongoing pain.

13. E · Treat as malignant spinal cord compression: catheterise, urgent senior/oncology discussion the same day for definitive treatment (chemotherapy ± steroids or surgery per oncologist), and urgent imaging if available

E is correct: progressive paraparesis, a sensory level and retention mean **spinal cord compression** (paraspinal/epidural Burkitt is stage III); function lost for more than 24–48 hours may not return, so this is a same-day oncology emergency. A is wrong: delay risks permanent paraplegia. B is wrong: LP can worsen compression (coning of the cord) and is not the urgent need. C is wrong: the neurological signs show cord involvement, not just bone pain. D is wrong: catheterising helps, but ibuprofen alone ignores the emergency.

14. B · Recognise this as treatment abandonment, contact the family, involve social work and support organisations for transport/lodging, re-counsel on curability, and plan to resume treatment

B is correct: missing **≥4 weeks** of planned curative therapy is **abandonment**, a leading cause of treatment failure in Uganda; it is addressed by tracing, social support and counselling, then resuming treatment. A is wrong: closing the file abandons the child. C is wrong: threats destroy trust and are untrue. D is wrong: herbal medicines are not curative and may be toxic. E is wrong: the oncologist decides the restart plan with the family after reassessment.

15. D · It is a B-cell lymphoma with a MYC translocation, nearly always EBV-positive, and frequently involves the jaw in young children

D is correct: endemic BL is a **B-cell** lymphoma with **MYC** translocation, nearly 100% **EBV-positive**, and classically presents with jaw tumours in young children. A is wrong: it is B-cell, and BCL2 translocation is typical of follicular lymphoma. B is wrong: almost all endemic cases are EBV-positive. C is wrong: peak age is 4–7 years (UCG). E is wrong: it is the fastest-growing human tumour (doubling ~25 h), so TLS is common.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The 'toothache' that would not go away

Opio, a 5-year-old boy from Lira, is referred to the regional referral hospital with a 1-month history of left jaw swelling. He had two teeth pulled at a private clinic with no improvement. For the past week his abdomen has been getting bigger.

1. What history do you take?

> Speed of growth (days–weeks), pain, loose teeth, eye swelling, abdominal pain/vomiting/constipation, **leg weakness, back pain, urinary retention** (cord compression), fever/night sweats/weight loss, **steroids or herbs** given, malaria history, **HIV status** (child and mother), immunisation, feeding, growth, distance from Kampala and family support.

2. Exam: weight 16 kg, height 108 cm (WAZ –1.6), pale, 7 × 6 cm firm left mandibular mass, loose molars, abdomen distended with a 10 cm irregular mass and shifting dullness, normal leg power, BP 100/60. What is your differential and leading diagnosis?

> Leading: **endemic Burkitt lymphoma, probable Murphy stage III** (jaw + extensive abdominal disease). Differentials: other NHL, rhabdomyosarcoma, dental abscess/osteomyelitis (no fever or pain), abdominal TB (ascites), Wilms/neuroblastoma (abdominal mass, but jaw involvement points to Burkitt).

3. Which investigations will you request?

> **FNA of the jaw mass** (and/or ascitic cytology): starry-sky cells. **FBC/film, LDH, uric acid, K^+, PO_4, Ca^{2+}, creatinine** (TLS risk). **HIV test**, malaria RDT. **Abdominal ultrasound** (kidneys, ovaries, nodes, ascites), **CXR**. **Bone marrow and CSF** at UCI for staging (stage IV if positive).

4. Results: LDH 1,700 U/L, uric acid 495 µmol/L, K^+ 4.9, PO_4 1.8, Ca^{2+} 2.2, creatinine 60 µmol/L. Before transfer, what do you start? (BSA ≈ 0.69 m²)

> **TLS prevention**: IV 0.9% saline (no potassium) at **2–3 L/m²/day ≈ 1.4–2 L/day** (≈60–85 mL/h), aiming for urine output **80–100 mL/m²/h (≈55–70 mL/h)**; **allopurinol 10 mg/kg/day ≈ 160 mg/day ÷ 3** (≈50 mg 8-hourly); weigh daily, chart input/output, repeat electrolytes 6–12-hourly. **No steroids.** Analgesia (paracetamol 15 mg/kg). Treat malaria if positive.

5. How will you refer and counsel the family?

> Same-week referral to the **UCI paediatric unit** with a clear note (history, sizes, results, treatment given, weight, height, BSA, HIV status). Explain that this is a **fast-growing but very treatable cancer**, that **chemotherapy at UCI is free**, and that finishing all cycles matters. Involve social work and child cancer support organisations for **transport, lodging and food**, to prevent abandonment.

6. At UCI, what treatment will he probably receive?

> A cyclophosphamide-based combination such as **COM** (cyclophosphamide, vincristine, methotrexate) with **intrathecal** chemotherapy, increasingly with **rituximab** (R-COM improved 1-year EFS from 43% to 67% in East Africa), alongside TLS prevention, nutrition support, and infection prophylaxis/treatment.

Take-away: A jaw mass in a young Ugandan child is Burkitt lymphoma until proven otherwise. Do not extract teeth or give steroids; get tissue, start TLS prevention and refer quickly with social support.

Case 2: A swollen tummy in a well toddler

Namutebi, 2 years 8 months, is brought to Mengo outpatients because her grandmother felt 'a hard ball' in her tummy while bathing her. She is eating and playing normally.

1. How do you examine the mass, and what must you avoid?

> Describe **site, size (measure), shape, surface, consistency, tenderness, mobility, movement with respiration, ballotability, whether it crosses the midline**, and check the **other kidney**, liver, spleen, nodes, genitalia (WAGR), aniridia, hemihypertrophy and **BP**. **Palpate gently and only once**, and ask others not to keep palpating (rupture risk).

2. Findings: 13 kg, smooth firm 10 × 8 cm right flank mass, ballotable, not crossing the midline; BP 122/78 (>95th centile for age); urine blood 1+. What is the leading diagnosis and the main differential?

> **Wilms tumour (nephroblastoma)**. Differentials: **neuroblastoma** (but she is well and the mass is smooth and does not cross the midline), hydronephrosis, multicystic dysplastic kidney, abdominal Burkitt, hepatoblastoma (liver origin).

3. What investigations are needed?

> **Abdominal ultrasound**: solid intrarenal mass (claw sign), check the **IVC and renal vein for tumour thrombus** and the **contralateral kidney** (bilateral = stage V). **CXR** (or CT chest) for lung metastases. FBC, U&E/creatinine, urinalysis. CT/MRI abdomen at the cancer centre. **No biopsy** for a typical case under SIOP-style care.

4. How is she treated in Uganda?

> Refer to **UCI**. SIOP-adapted approach: **pre-operative chemotherapy** (vincristine + actinomycin-D for about 4 weeks for localised disease; add doxorubicin if metastatic) → **nephrectomy** by a paediatric surgeon → **post-operative chemotherapy** (± radiotherapy) according to stage and histology. Treat hypertension (e.g. amlodipine 0.1–0.2 mg/kg/day, max 10 mg).

5. What are the parents likely to ask, and how do you answer?

> 'Will she lose her kidney?' Usually yes, one kidney is removed; **one healthy kidney is enough** for normal life, with BP and urine checks later. 'Can it be cured?' **Most Wilms tumours are curable**, especially when treated early and completely. 'Did we cause it?' No; it arises from embryonic kidney cells.

6. Present her on the ward round.

> "Namutebi, 2 years 8 months, 13 kg (WAZ −0.6), fully immunised, eating family foods, HIV-unexposed, presents with an incidentally found smooth right flank mass not crossing the midline, with hypertension and microscopic haematuria. Ultrasound shows a solid right renal mass with a normal IVC and left kidney; CXR clear. Probable unilateral Wilms tumour. Plan: urgent UCI referral for pre-operative chemotherapy then nephrectomy; antihypertensive; minimal palpation."

> **Take-away:** A well toddler with a smooth flank mass that does not cross the midline is Wilms until proven otherwise. Palpate once, check BP and the other kidney, image the IVC and chest, avoid local surgery or biopsy, and refer for SIOP-style care.

Case 3: The child with 'malaria that won't go'

Ssekandi, a 4-year-old boy, is admitted to a HC IV in Masaka with fever for 3 weeks despite two courses of antimalarials. His mother says he refuses to walk because his legs hurt and he bruises easily.

1. What clues point away from malaria?

> Fever **>2 weeks** despite treatment, **bone pain/limp**, **bruising** (thrombocytopenia), progressive pallor. Look for **lymphadenopathy, hepatosplenomegaly, petechiae, gum bleeding, testicular swelling**, and check the RDT (and film).

2. Exam: 14 kg, pale, petechiae on legs, cervical nodes 2 cm, spleen 5 cm, liver 4 cm. RDT negative. HemoCue Hb 5.8 g/dL. FBC: WBC 62 × 10^9/L, platelets 18 × 10^9/L, 'many atypical cells'. What is your working diagnosis?

> **Acute leukaemia, most likely ALL** (age, organomegaly, nodes, bone pain), with anaemia, thrombocytopenia and high WBC. Differential: AML, overwhelming infection, visceral leishmaniasis (wrong region), SCD crisis (not explained by blasts).

3. What will you do before transfer and what must you avoid?

> Do: **packed cells 10 mL/kg slowly** if symptomatic (care with high WBC), **platelets** if bleeding and available, **IV fluids without potassium** + **allopurinol 10 mg/kg/day** (TLS), **blood culture and IV antibiotics** if febrile (assume neutropenia), paracetamol for pain. **Avoid**: **steroids**, aspirin/NSAIDs (bleeding), IM injections, and a rapid large transfusion (hyperviscosity).

4. What will be done at UCI to confirm and classify?

> Peripheral film, **bone marrow aspirate**, **flow cytometry** (B- vs T-ALL vs AML), cytogenetics if available, **lumbar puncture** (CNS involvement), CXR (mediastinal mass), uric acid/LDH/electrolytes, HIV test, hepatitis B, echo before anthracyclines.

5. Outline the treatment and how long it lasts.

> Risk-adapted multiphase chemotherapy over **about 2–2.5 years**: **induction** (steroid, vincristine, asparaginase ± anthracycline + intrathecal methotrexate), **consolidation/CNS therapy**, **delayed intensification**, and **maintenance** (daily oral mercaptopurine, weekly methotrexate, monthly vincristine/steroid pulses). Plus cotrimoxazole PCP prophylaxis, nutrition support and infection management.

6. What will you tell the family about the outlook?

> ALL is **the most curable childhood leukaemia** (about 90% in high-income settings, and improving in Uganda). The biggest risks are infection during treatment and stopping treatment early. They should come immediately for fever, bleeding or lethargy, and complete every phase.

> **Take-away:** Persistent fever, bone pain, bruising and organomegaly with a negative RDT mean an FBC and film, not another antimalarial course. Avoid steroids, prevent TLS and infection, and refer urgently.

References

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.